Ch. 5 Computation

1 Programs

1.1 Logical Instructions

Define $\mathcal{L}$; an ordered set of logical operations s_i

$$\mathcal{L} := \{s_1, s_2, ..., s_N\}$$

1.2 Memory

Define Memory $\mathcal{M}$; an ordered set of elements, magnitudes, or sets

$$\mathcal{M} := \{b_1, b_2, ..., b_M\}$$

1.3 State |

Define state; the memory utilized to perform program P

$$P := \{s_1, s_2, ..., s_N | b_1, b_2, ..., b_M\} =$$
$$\{s_1, s_2, ..., s_N, b_1, b_2, ..., b_M\}$$

1.4 Boolean Programs

Define a boolean program; boolean programs can represent functions with inputs x_i, input set C, and boolean output y_o

$$X = \{x_1, ..., x_n, C\}; \;\; C = \{u_1, u_2, ..., u_c\}$$
$$P = P[X] := \{s_1, s_2, ..., s_N \mid b_1, b_2, ..., b_M, y_o\} =$$
$$P[X] \to y_o \in \{\mathbb{T}, \mathbb{F}\}$$

1.5 Void Programs

Define a void program; a program with inputs x_i, input set C, and no output

$$X = \{x_1, ..., x_n, C\}$$
$$P = P[X] := \{s_1, s_2, ..., s_N \mid b_1, b_2, ..., b_M\}$$

1.6 Numerical Programs

Define a numerical program; a program with inputs x_i, input set C, and real, rational output y_o

$$X = \{x_1, ..., x_n, C\}$$
$$P = P[X] := \{s_1, s_2, ..., s_N \mid b_1, b_2, ..., b_M, y_o\} =$$
$$P[X] \to y_o \in \mathbb{Q} \;\; y_o \geqslant 0$$

1.7 System Programs

Define a system program; a program with inputs x_i, input set C, and real, output set Y_o

$$X = \{x_1, ..., x_n, C\}$$
$$P = P[X] := \{s_1, s_2, ..., s_N \mid b_1, b_2, ..., b_M, Y_o\} =$$
$$P[X] \to Y_o = \{y_1, y_2, ..., y_K\}$$

1.8 Mathematical Programs

Define a mathematical program; a program with inputs x_i, input set C and numerical output y_o

$$X = \{x_1, ..., x_n, C\}$$
$$P = P[X] := \{s_1, s_2, ..., s_N \mid b_1, b_2, ..., b_M, y_o\} =$$
$$P[X] \to y_o \in \mathbb{Q}$$

2 No-op ;

2.1 Definition

$$; := \varnothing$$

2.2 Property of No-op

No-op can be inserted into any set with equality

$$S = \{s_1, s_2, ..., s_N\}$$
$$S_; = insert[S, ;, i]$$
$$S_; = S_1 \ \ \forall i$$
$$|S_;| = |S| \ \ \forall i$$

2.3 Proof

by definition of magnitude of null = 0 with Set And

3 Decision Problems

3.1 Definition

Define decision problem; a function with inputs x_i and boolean output "answer" a_o

$$X_i = \{x_1, ..., x_n, C\}$$
$$D := f[X_i] \rightarrow a_o \in \{\mathbb{T}, \mathbb{F}\} \quad \forall X_i$$

4 Solutions

4.1 Definition

Program P is a solution s^+ to decision problem D if

1. P outputs answer a_o for all inputs $X_i \ \ \forall i$
and
2. $s^+[X_i]$ is a subset of $s^+[\hat{X}_i]$

$$X_i = \{x_1, ..., x_n, C\}; \ \ \hat{X}_i = \{x_1, ..., x_n, x_{n+1}, C\}$$
$$D := f[X_i] \rightarrow a_o \in \{\mathbb{T}, \mathbb{F}\} \quad \forall X_i$$
$$s^+ = s^+[X_i] := P :$$
$$(P[X_i] \rightarrow y_o == a_o \quad \forall X_i) \ \ \cap \ \ (P[\hat{X}_i] \supseteq P[X_i] \quad \forall X_i, \hat{X}_i)$$
$$P[X_i] = \{s_1, s_2, ..., s_N | b_1, b_2, ..., b_M, y_o\}$$
$$s^+ = P[X_i] = \{s_1, s_2, ..., s_{O_T[n]}, b_1, b_2, ..., b_{O_S[n]}, y_o\} \quad \forall X_i$$

4.1.1 Property of No-op ;

No-op ; can be added to any solution S_i without modifying the output y_o

$$s^+ = \{s_1, s_2, ..., s_{O_T[n]}, b_1, b_2, ..., b_{O_S[n]}, y_o\}$$
$$\hat{s}^+ \rightarrow \hat{y}_o = insert[s^+, ;, k]$$
$$\hat{y}_o = y_o \ \ \forall k$$

4.2 Definition of S^+

Define S^+; the set of solutions to decision problem D

$$X_i = \{x_1, ..., x_n, C\}; \;\; \hat{X}_i = \{x_1, ..., x_n, x_{n+1}, C\}$$
$$D := f[X_i] \rightarrow a_o \in \{\mathbb{T}, \mathbb{F}\} \quad \forall X_i$$
$$s_j^+ = s_j^+[X_i] := P :$$
$$(P[X_i] \rightarrow y_o == a_o \quad \forall X_i) \;\; \cap \;\; (P[\hat{X}_i] \supseteq P[X_i] \quad \forall X_i, \hat{X}_i)$$
$$S^+ := \{s_j^+, ...\} \quad \forall j$$

4.3 Definition of Solvable

Define solvable

$$X_i = \{x_1, ..., x_n, C\}; \;\; \hat{X}_i = \{x_1, ..., x_n, x_{n+1}, C\}$$
$$D := f[X_i] \rightarrow a_o \in \{\mathbb{T}, \mathbb{F}\} \quad \forall X_i$$
$$solvable := solvable[D] \rightarrow b_o \in \{\mathbb{T}, \mathbb{F}\} =$$
$$\exists P : (P[X_i] \rightarrow y_o == a_o \quad \forall X_i) \;\; \cap \;\; (P[\hat{X}_i] \supseteq P[X_i] \quad \forall X_i, \hat{X}_i)$$

5 The set of all Decision Problems $\mathbb{D}$

5.1 Definition

Define the set of decision problems $\mathbb{D}$

$$X_i = \{x_1, ..., x_n, C\}$$
$$D_j := f_j[X_i] \rightarrow a_o \in \{\mathbb{T}, \mathbb{F}\} \quad \forall X_i$$
$$\mathbb{D} := \{D_j, ...\} \quad \forall j$$

6 Instruction and Memory Notation

Define $\mathcal{L}$ a set of logical operations
Define $\mathcal{M}$ a set of memory elements, magnitudes, and sets

$$X_i = \{x_1, ..., x_n, C\};$$
$$P[X_i] \rightarrow y_o = \{s_1, s_2, ..., s_{O_T[n]}, b_1, b_2, ..., b_{O_S[n]}, y_o\}$$
$$\mathcal{L} := \{s_1, s_2, ..., s_{O_T[n]}\}$$
$$\mathcal{M} := \{b_1, b_2, ..., b_{O_S[n]}\}$$
$$P[X_i] = \{\mathcal{L}, \mathcal{M}, y_o\}$$

7 Complexity

7.1 Time Complexity of a Decision Problem $O_T[n]$

Define Time Complexity $O_T[n]$ of solution s^+ to Decision Problem D as the total number of logical operations

$$X_i = \{x_1, ..., x_n, C\}; \;\; \hat{X}_i = \{x_1, ..., x_{n+1}, C\}$$

$$D := f[X_i] \rightarrow a_o \in \{\mathbb{T}, \mathbb{F}\} \quad \forall X_i$$

$$s^+[X_i] := P :$$
$$(P[X_i] \rightarrow y_o == a_o \quad \forall X_i) \;\; \cap \;\; (P[\hat{X}_i] \supseteq P[X_i] \quad \forall X_i, \hat{X}_i)$$

$$s^+ = \{s_1, s_2, ..., s_N | b_1, b_2, ..., b_M\} = \{s_1, s_2, ..., s_{O_T[n]}, b_1, b_2, ..., b_{O_S[n]}, y_o\}$$
$$= \{\mathcal{L}, \mathcal{M}, y_o\}$$

$$O_T[n] := |\mathcal{L}| = N$$

7.2 Space Complexity $O_S[n]$

Define Space Complexity $O_S[n]$ of solution s^+ to Decision Problem D as the total number of memory elements

$$X_i = \{x_1, ..., x_n, C\}; \;\; \hat{X}_i = \{x_1, ..., x_{n+1}, C\}$$

$$D := f[X_i] \rightarrow a_o \in \{\mathbb{T}, \mathbb{F}\} \quad \forall X_i$$

$$s^+[X_i] := P :$$
$$(P[X_i] \rightarrow y_o == a_o \quad \forall X_i) \;\; \cap \;\; (P[\hat{X}_i] \supseteq P[X_i] \quad \forall X_i, \hat{X}_i)$$

$$s^+ = \{s_1, s_2, ..., s_N | b_1, b_2, ..., b_M\} = \{s_1, s_2, ..., s_{O_T[n]}, b_1, b_2, ..., b_{O_S[n]}, y_o\}$$
$$= \{\mathcal{L}, \mathcal{M}, y_o\}$$

$$O_S[n] := |\mathcal{M}| + |y_o|^* = M + 1$$

*It is convention to reserve one memory element for output y_o.
Void programs do not require the y_o memory element for output

8 Definition of Complexity

Define Complexity $O[n]$ as a vector of dimension Y

$$\mathbf{O}[n] := \ < O_T[n], O_S[n], O_3[n], O_4[n]..., O_V[n] >$$

9 Total Complexity

$$O[n] := O_T[n] + O_S[n] + \sum_{i=3}^{V} O_i[n]$$

10 Simple Computational Complexity

The remainder of this chapter assumes simple computational complexity of dimension 2

10.1 Definition

Define simple computational complexity of dimension 2

$$\mathbf{O}[n] := \ < O_T[n], O_S[n] >$$

10.2 Time Complexity

Restate definition of Time Complexity $O_T[n]$ of solution s^+

$$s^+ = \{\mathcal{L}, \mathcal{M}, y_o\}$$

$$O_T[n] := |\mathcal{L}| = N$$

10.3 Space Complexity

Restate definition of Time Complexity $O_S[n]$ of solution s^+

$$s^+ = \{\mathcal{L}, \mathcal{M}, y_o\}$$

$$O_S[n] := |\mathcal{M}| + |y_o| = M + 1$$

10.4 Total Complexity

$$O[n] := O_T[n] + O_S[n]$$

$$= |\mathcal{L}| + |\mathcal{M}| + |y_o| = N + M + 1$$

10.5 $O_S[n] \neq 0$

10.5.1 Proof

By definition of decision problem; Proof by contradiction; y_o must be set to TF by definition; Suppose yo = 0; then yo is empty set; contradicts definition of D

10.6 $O_T[n] \neq 0$

10.6.1 Proof

By definition of decision problem; Proof by contradiction; y_o must be set to TF by definition; Suppose |L| = 0; yo <- TF cap L is null by definition of empty set; implies yo emptyset (doesnt exist)

10.7 $\mathbf{O[n]} = O_T[n] + O_S[n] \neq \mathbf{0}$

10.7.1 Proof

10.8 $O[n] > O_T[n]$

10.8.1 Proof

10.9 $O[n] > O_S[n]$

10.9.1 Proof

10.10 $O[n+1] \geqslant O[n]$

11 Optimal Complexity

11.1 Definition

Define Optimal Complexity; the minimum total complexity required to solve a decision problem

$$O_{opt}[n] := O[n] :$$
$$\nexists O_i[n] < O[n] \quad \forall s_i \in S^+$$

11.2 Proof of Existence

Prove the existence of at least one $O_{min}[n]$ by induction/contradiction
Induction, let n = 1
There must exists an O_{min}
subfunction property all the way up

12 Optimal solution

Define an optimal solution s^+_{opt}

12.1 Definition

$$X_i = \{x_1, ..., x_n\}$$
$$D_j := f[X_i] \to a_o \in \{\mathbb{T}, \mathbb{F}\} \quad \forall X_i$$
$$s^+ := P[X_i] \to y_o : y_o = a_o \quad \forall X_i$$
$$s^+_{opt} := s^+ :$$
$$\nexists \hat{O}[n] < O_{opt}[n] \quad \forall n,\ s^+ \in S^+_j$$

12.2 Optimal Time Complexity Solution

$$X_i = \{x_1, ..., x_n\}$$
$$D_j := f[X_i] \to a_o \in \{\mathbb{T}, \mathbb{F}\} \quad \forall X_i$$
$$s^+ := P[X_i] \to y_o : y_o = a_o \quad \forall X_i =$$
$$\{s_1, s_2, ..., s_{O_T[n]}, b_1, b_2, ..., b_{O_S[n]}, X_i, y_o\} = \{\mathcal{L}, \mathcal{M}, X_i, y_o\}$$
$$O_T[n] := |\mathcal{L}| = N$$
$$s^+_T := s^+ :$$

$$\nexists \hat{O_T}[n] < O_T[n] \quad \forall n,\ s^+ \in S_j^+$$

12.3 Optimal Space Complexity Solution

$$X_i = \{x_1, ..., x_n\}$$

$$D_j := f[X_i] \rightarrow a_o \in \{\mathbb{T}, \mathbb{F}\} \quad \forall X_i$$

$$s^+ := P[X_i] \rightarrow y_o : y_o = a_o \quad \forall X_i =$$

$$\{s_1, s_2, ..., s_{O_T[n]}, b_1, b_2, ..., b_{O_S[n]}, X_i, y_o\} = \{\mathcal{L}, \mathcal{M}, X_i, y_o\}$$

$$O_S[n] := |\mathcal{M}| = M$$

$$s_S^+ := s^+ :$$

$$\nexists \hat{O_S}[n] < O_S[n] \quad \forall n,\ s^+ \in S_j^+$$

12.4 Conjecture of Optimal Solutions

$O_{T_{min}}$ subject to $O_{S_{opt}} = 1$

$$\exists s_{opt}^+ :$$

$$O_{opt}[n] = 1 + O_T[n] = O_{S_{opt}} + O_T[n] \quad \forall s^+ \in S^+$$

12.4.1 Proof

12.5 Theorem of Optimal Solutions

Alternate way of expressing $O_{opt}[n]$ possibly with efficiency function and equivalence functions?
$f_{S \rightarrow T}[n, O_S[n]]$ as a function of space complexity order K?
Efficiency function ($f_{S \rightarrow T}[n, O_S[n]]$ as a function of space complexity order K) is strictly decreasing for Polynomial Functions
Or there's an inflection point
Efficiency function might have a general pattern for all problems in D

13 Polynomial Complexity

13.1 Definition

Decision problem D with solution s^+ has polynomial total complexity $O[n]$ if

$$\exists K, C, \lambda_1 ... \lambda_K \quad :$$
$$O[n] = (\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C \quad \forall n$$

13.2 Polynomial Problems

Define $\mathbb{P}$, the set of Decision Problems that can be solved with Polynomial Complexity

$$\mathbb{P} := \{D_1, D_2, ...\} :$$
$$\exists K, C, \lambda_1 ... \lambda_K :$$
$$O[n] = (\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C \quad \forall n, D_i \in \mathbb{P}$$

13.3 Polynomial Order of Complexity

Total complexity $O[n]$ is said to be of order K_{opt}

$$O[n] \sim K_{opt}$$
$$O_{opt}[n] := O[n] :$$
$$\nexists \hat{O}[n] < O_{opt}[n] \ \ \forall n$$
$$O_{opt}[n] < (\lambda_{K_{opt}} n)^{K_{opt}} + (\lambda_{K_{opt}-1} n)^{K_{opt}-1} ... + \lambda_1 n + C \quad \forall n$$
$$K_{opt} := K :$$
$$\nexists \hat{K} : O_T[n] < (\lambda_{\hat{K}} n)^{\hat{K}} + (\lambda_{\hat{K}-1} n)^{\hat{K}-1} ... + \lambda_1 n + C \quad \forall n, \ \hat{K} < K$$

13.4 Corrolary of Optimal Complexity

$$\nexists s^+ \in S^+ :$$
$$O_T[n] < (\lambda_{\hat{K}} n)^{\hat{K}} + (\lambda_{\hat{K}-1} n)^{\hat{K}-1} ... + \lambda_1 n + C \quad \forall n, \ \hat{K} < K_{opt}$$

13.4.1 Proof

Proof by contradiction; definition of optimal complexity

13.5 Property of Polynomial Complexity 1

$$lim_{n\to\infty}\frac{O[n+1]}{O[n]} = 1$$

13.5.1 Proof WIP

FIX!!! Show there exists no constant satisfying the decreasing limit condition

$$O[n] < (\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C$$
$$O[n+1] < (\lambda_K (n+1))^K + (\lambda_{K-1}(n+1))^{K-1} ... + \lambda_1 (n+1) + C$$
$$O[n] \sim (\lambda n)^K;\ \ O[n+1] \sim (\lambda n)^K$$
$$lim_{n\to\infty}\frac{(\lambda n)^K}{(\lambda n)^K} = 1$$

13.6 Property of Polynomial Complexity 2

$$\exists K, C, \lambda_1, ..., \lambda_K :$$
$$(O[n+1] - O[n]) = f_{n+1}[n] = (\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C \quad \forall n$$

13.6.1 Proof FIX!!!

$$O[n+1] < (\lambda_K (n+1))^K + (\lambda_{K-1}(n+1))^{K-1} ... + \lambda_1 (n+1) + C$$
$$O[n] < (\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C$$

13.7 Total Polynomial Complexity Implies Time bounded Polynomial Complexity

$$D \in \mathbb{P} \Longrightarrow O_T[n] < ...$$

13.7.1 Proof FIX!!!

$$O[n] < (\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C \ \ \forall n$$
$$O[n] := O_T[n] + O_S[n];\ \ O_T[n] < O[n]$$
$$\therefore O_T[n] < (\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C \ \ \forall n$$

13.8 Total Polynomial Complexity Implies Space bounded Polynomial Complexity

$$D \in \mathbb{P} \Longrightarrow O_S[n] < ...$$

13.8.1 Proof FIX!!!

$$O[n] < (\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C \;\; \forall n$$
$$O[n] := O_T[n] + O_S[n]; \;\; O_S[n] < O[n]$$
$$\therefore O_S[n] < (\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C \;\; \forall n$$

13.9 Total Polynomial Complexity iff Time and Space bounded by Polynomial Complexity

Use limit definition

13.10 Order of Complexity

ERROR in second condition
Total Complexity is said to be on the order of K_{max}

$$O[n] \sim K_{max}$$
$$K_{max} := K :$$
$$O[n] < (\lambda_{K_{max}} n)^{K_{max}} + (\lambda_{K_{max}-1} n)^{K_{max}-1} ... + \lambda_1 n + C \quad \forall n$$
$$\nexists O[n] < (\lambda_{\hat{K}_{max}} n)^{\hat{K}_{max}} + (\lambda_{\hat{K}_{max}-1} n)^{\hat{K}_{max}-1} ... + \lambda_1 n + C \quad \forall n, \hat{K} < K_{max}$$

13.11 Theorem Either OT or OS is on the order of Oopt

Proof by contradiction

14 Polynomial Time Complexity

14.1 Definition

Decision problem D with (optimal) Time Complexity $O_T[n]$ is bounded by polynomial time complexity if

$$\exists K, C, \lambda_1 ... \lambda_K \quad :$$
$$O_T[n] < (\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C \quad \forall n$$

14.2 Polynomial Time Solutions

Define $\mathbb{S}^+_{time}$, the set of solutions that can be solved with polynomial time complexity

$$\mathbb{S}^+_{time} := \{s^+_1, s^+_2, ...\} :$$
$$\exists K, C, \lambda_1 ... \lambda_K :$$
$$O_T[n] < (\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C \quad \forall n, s_i \in \mathbb{S}^+_{time}$$

14.3 Property of Polynomial Time Complexity 1

$$lim_{n \to \infty} \frac{O_T[n+1]}{O_T[n]} = 1$$

14.3.1 Proof

14.4 Property of Polynomial Time Complexity 2

$$\exists K, C, \lambda_1, ..., \lambda_K :$$
$$(O_T[n+1] - O_T[n]) < (\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C \ \ \forall n$$

14.4.1 Proof

14.5 Order of Complexity

Time complexity $O_T[n]$ is said to be on the order of K_{max}

$$O_T[n] < (\lambda_{K_{max}} n)^{K_{max}} + (\lambda_{K_{max}-1} n)^{K_{max}-1} ... + \lambda_1 n + C$$
$$O_T[n] \sim K_{max}$$

14.6 Proof of the existence of $O_{T_{opt}}$

15 Polynomial Space Complexity

15.1 Defintion

Decision problem D with (optimal) Time Complexity $O_S[n]$ is bounded by polynomial time complexity if

$$\exists K, C, \lambda_1 ... \lambda_K \quad :$$
$$O_S[n] < (\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C \quad \forall n$$

15.2 Polynomial Space Problems

Define $\mathbb{S}^+_{space}$, the set of solutions that can be solved with polynomial time complexity

$$\mathbb{S}^+_{space} := \{s_1, s_2, ...\} :$$
$$\exists K, C, \lambda_1 ... \lambda_K :$$
$$O_S[n] < (\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C \quad \forall n, s_i \in \mathbb{S}^+_{time}$$

15.3 Total Polynomial Complexity Implies Space bounded Polynomial Complexity

15.4 Space Bounded Polynomial Complexity Implies Total Polynomial Complexity

15.5 Polynomial Space Complexity iff Polynomial Complexity

15.6 Property of Polynomial Space Complexity 1

$$lim_{n \to \infty} \frac{O_S[n+1]}{O_S[n]} = 1$$

15.6.1 Proof

15.7 Property of Polynomial Space Complexity 2

$$\exists K, C, \lambda_1, ..., \lambda_K :$$
$$(O_S[n+1] - O_S[n]) < (\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C \ \ \forall n \ \ \forall n$$

15.7.1 Proof

15.8 Order of Complexity

Space complexity $O_S[n]$ is said to be on the order of K_{max}

$$O_S[n] < (\lambda_{K_{max}} n)^{K_{max}} + (\lambda_{K_{max}-1} n)^{K_{max}-1} ... + \lambda_1 n + C$$
$$O_S[n] \sim K_{max}$$

15.9 Proof of the existence of $O_{S_{opt}}$

16 Inductive Functions

16.1 Inductive Function f_{n+1}

$$O[n] := O_T[n] + O_S[n]$$
$$O[n+1] = O_T[n+1] + O_S[n+1]$$

$$f_{n+1}[n] := f[n] :$$
$$O[n+1] = f[n] + O[n] \quad \forall n$$

16.1.1 Proof of existence

Algebraic Proof

16.2 Inductive Space and Time Formulas

$$f^T_{n+1} := O_T[n+1] - O_T[n]$$
$$O_T[n+1] = O_T[n] + f^T_{n+1}$$
$$f^S_{n+1} := O_S[n+1] - O_S[n]$$
$$O_S[n+1] = O_S[n] + f^S_{n+1}$$

16.2.1 Proof of existence

Algebraic Proof

16.3 Inductive Function Expressions

Relate $f_{n+1}[n]$ to equivalence functions

$$D \in \mathbb{P}$$
$$O[n] := O_T[n] + O_S[n]$$
$$O[n+1] = O_T[n+1] + O_S[n+1] = O[n] + f_{n+1}[n]$$
$$O_T[n] = O[n] - O_S[n]$$
$$O_S[n] = O[n] - O_T[n]$$

$$f_{n+1} = O[n+1] - O[n]$$
$$f_{n+1} = O_T[n+1] + O_S[n+1] - O[n]$$

$$f_{n+1} = O_T[n+1] - O_T[n] + O_S[n+1] - O_S[n]$$
$$f_{n+1} = O[n+1] - O_T[n] - O_S[n]$$
$$f_{n+1}[n] = f^T_{n+1}[n] + f^S_{n+1}[n]$$

16.4 Zero Order Inductive Function

$$Let\ \ O_S[n] \sim n^0$$
$$f_{n+1} = O_T[n+1] - O_T[n] + O_S[n+1] - O_S[n] = O_T[n+1] - O_T[n]$$

16.5 Property of Polynomial Complexity

$f_{n+1}[n]$ has order less than O[n]
$f_{n+1}[n]$ is bound by K_{max} - 1

16.5.1 Proof

Proof by contradiction; limit doesn't converge

17 Duality of $O_T[n]$, $O_S[n]$?

17.0.1 O_T to O_S

Define equivalence function $f_{T\to S}$; a function converting logical operations into memory elements

$$f_{T\to S} := f :$$
$$O_S[n] = f[n, O_T[n]] \quad \forall n, s^+ \in S^+$$

17.0.2 O_S to O_T

Define equivalence function $f_{S\to T}$; a function converting memeory elements into logical operations

$$f_{S\to T} := f :$$
$$O_T[n] = f[n, O_S[n]] \quad \forall n, s^+ \in S^+$$

17.0.3 Invertibility?

17.0.4 Polynomial Bounded?

17.1 Efficiency Function?

Function relating the decrease in O[n] as $O_S[n]$ increases in order $f_{S\to T}[n, O_S[n]]$ as a function of space complexity order K?

18 Theorem of Computational Duality?

For all Problems in P there exists a duality function
Formally define dynamic programming, Optimal polynomial complexity minimizes the difference between time and space complexity order

$$D \in \mathbb{P}$$
$$O[n] := O_T[n] + O_S[n]$$
$$limit_{n\to\infty} \frac{O[n+1]}{O[n]} = 1 \quad \forall s^+ \in S^+_{\mathbb{P}}$$

$$O_T[n] = f_{S\to T}[n, O_S[n]]$$
$$O_S[n] = f_{T\to S}[n, O_T[n]]$$

$$limit_{n\to\infty} \frac{O_T[n+1]+O_S[n+1]}{O_T[n]+O_S[n]} = 1 \quad \forall s^+ \in S^+_{\mathbb{P}}$$

$$limit_{n\to\infty} \frac{f_{S\to T}[n+1,O_S[n+1]]+O_S[n+1]}{f_{S\to T}[n,O_S[n]]+O_S[n]} = 1 \quad \forall s^+ \in S^+_{\mathbb{P}}$$

$$limit_{n\to\infty} \frac{O_T[n+1]+f_{T\to S}[n+1,O_T[n+1]]}{O_T[n]+f_{T\to S}[n,O_T[n]]} = 1 \quad \forall s^+ \in S^+_{\mathbb{P}}$$

19 Subfunctions

19.1 Restate the subfunction condition of solutions

19.2 Show O[n] is a strictly increasing function

19.3 Theorem of Polynomial Subfunctions

19.4 Theorem of Divergent Subfunctions

20 Sum to N Problem with 2 integers

20.1 State formal definition of Sum to N : $x_i + x_j == N$

$$X_i = \{x_1, ..., x_n, N\}$$
$$D := f[X_i] \rightarrow a_o \in \{\mathbb{T}, \mathbb{F}\} \quad \forall X_i$$
$$s^+ = P :$$
$$(P[X_i] \rightarrow y_o == a_o \quad \forall X_i) \quad \cap \quad (P[\hat{X}_i] \supseteq P[X_i] \quad \forall X_i, \hat{X}_i)$$
$$s^+ = \{s_1, s_2, ..., s_{O_T[n]}, b_1, b_2, ..., b_{O_S[n]}, y_o\} = \{\mathcal{L}, \mathcal{M}, y_o\}$$

$$D = f[X_i] = \exists x_j, x_k \in X_i : x_j + x_k == N$$

20.2 Express a formal solution : $O_S[n] \sim n^0$

$$s^+ = \{s_1, s_2, ..., s_{O_T[n]}, b_1, b_2, ..., b_{O_S[n]}, y_o\} = \{\mathcal{L}, \mathcal{M}, y_o\}$$
$$s_1 = y_o \leftarrow \mathbb{F};$$
$$\forall i, j > i$$
$$s_2, s_3, s_8, s_9, ..., s_{3ij-4}, s_{3ij-3}..., s_{3n(n-1)-4}, s_{3n(n-1)-3} = b_1 \leftarrow x_i + x_j$$
$$s_4, s_5, s_{10}, s_{11}, ..., s_{3ij-2}, s_{3ij-1}..., s_{3n(n-1)-2}, s_{3n(n-1)-1} = b_1 \leftarrow b_1 == N$$
$$s_6, s_7, s_{12}, s_{13}..., s_{3ij}, s_{3ij+1}..., s_{3n(n-1)}, s_{3n(n-1)+1} = y_o \leftarrow y_o \vee b_1$$
$$s^+ = \{y_o \leftarrow \mathbb{F}, y_o \leftarrow y_o \vee (x_i + x_j == N) \quad \forall i, j > i \mid b_1, y_o\}$$

20.3 Show s^+ satisfies the subfunction condition of solutions: $P[\hat{X}_i] \supseteq P[X_i] \quad \forall \hat{X}_i, X_i$

$$X_i = \{x_1, ..., x_n, N\}; \quad \hat{X}_i = \{x_1, ..., x_n, x_{n+1}, N\}$$
$$s^+ = \{s_1, s_2, ..., s_{O_T[n]}, b_1, b_2, ..., b_{O_S[n]}, y_o\} = \{\mathcal{L}, \mathcal{M}, y_o\}$$
$$s^+_{n+1} = s^+ \cup \hat{s}^+$$
$$s_1 = y_o \leftarrow \mathbb{F};$$
$$\forall i, j > i$$
$$s_2, s_3, s_8, s_9, ..., s_{3ij-4}, s_{3ij-3}..., s_{3n(n-1)-4}, s_{3n(n-1)-3} = b_1 \leftarrow x_i + x_j$$
$$s_4, s_5, s_{10}, s_{11}, ..., s_{3ij-2}, s_{3ij-1}..., s_{3n(n-1)-2}, s_{3n(n-1)-1} = b_1 \leftarrow b_1 == N$$

$$s_6, s_7, s_{12}, s_{13}..., s_{3ij}, s_{3ij+1}..., s_{3n(n-1)}, s_{3n(n-1)+1} = y_o \leftarrow y_o \vee b_1$$

$$\forall k$$

$$s... = b_1 \leftarrow x_k + x_{n+1}$$

$$s... = b_1 \leftarrow b_1 == N$$

$$s... = y_o \leftarrow y_o \vee b_1$$

$$s^+ = \{y_o \leftarrow \mathbb{F}, y_o \leftarrow y_o \vee (x_i + x_j == N) \quad \forall i, j > i \mid b_1, y_o\}$$

$$\hat{s}^+ = \{y_o \leftarrow y_o \vee (x_k + x_{n+1} == N) \ \ \forall k < n+1 | \, b_1, y_o\}$$

$$s^+_{n+1} = \{y_o \leftarrow \mathbb{F}, y_o \leftarrow y_o \vee (x_i + x_j == N) \quad \forall i, j > i \mid b_1, y_o\} \ \cup$$

$$\{y_o \leftarrow y_o \vee (x_k + x_{n+1} == N) \ \ \forall k < n+1 | \, b_1, y_o\}$$

$$s^+_{n+1} = P[\hat{X}_i] \supseteq P[X_i] = s^+$$

20.4 Determine $O[n], O_S[n], O_T[n], \hat{O}[n], \hat{O}_T[n], \hat{O}_S[n]$ for the above solution

$$O_S[n] = |y_o| + |b_1| = 2$$

$$O_T[n] = 3n(n-1) + 1 = 3n(n-1) - 1 + O_S[n]$$

$$O[n] = 3n(n-1) + 3 = 3n^2 - 3n + 3$$

$$\hat{O}_S[n] = 0$$

$$\hat{O}_T[n] = 6n$$

$$\hat{O}[n] = \hat{O}_S[n] + \hat{O}_T[n]$$

20.5 Verify $O[n+1] = O[n] + \hat{O}[n]$

$$O[n+1] = O[n] + \hat{O}[n]$$

$$3(n+1)^2 - 3(n+1) + 3 = 3n^2 - 3n + 3 + 6n$$

$$3n^2 + 6n + 3 - 3n - 3 + 3 = 3n^2 + 3n + 3$$

$$3n^2 + 3n + 3 = 3n^2 + 3n + 3$$

20.6 Show s^+ has Polynomial Complexity by the definition of Total Polynomial Complexity

$$O[n] = 3n^2 - 3n + 3$$

20.7 Show s^+ has Polynomial Complexity by showing limit$_{n\to\infty} \frac{O[n+1]}{O[n]}$ = 1

$$limit_{n\to\infty} \frac{O[n+1]}{O[n]} =$$

$$limit_{n\to\infty} \frac{3n^2+3n+3}{3n^2-3n+3} =$$

$$limit_{n\to\infty} (\frac{3n^2-3n+3}{3n^2-3n+3} + \frac{6n}{3n^2-3n+3}) =$$

$$limit_{n\to\infty} (1 + \frac{6n}{3n^2-3n+3}) = 1$$

21 Divergent Problems

21.1 Definition

$$\hat{\mathcal{D}} := \{\hat{D}_1, \hat{D}_2, ...\} :$$
$$lim_{n\to\infty} \frac{O[n+1]}{O[n]} \;\; diverges \;\;\; \forall \hat{D} \in \hat{\mathcal{D}}$$

21.2 Theorem of Divergent Subfunctions

If an (inductive) subfunction of s^+ diverges, the solution is divergent

$$f_{n+1} = \sum g_{n+1}$$
$$limit \frac{g[n+1]}{g[n]} diverges$$

$$\exists g_{n+1} : limit \frac{g_{n+1}}{O[n]} diverges \Longrightarrow limit \frac{O[n+1]}{O[n]} diverges$$

21.3 Proof

21.4 The Set of Polynomial Solutions and the Set of Divergent Solutions are disjoint

$$\mathbb{P} \cap \hat{D} = \varnothing$$

21.5 Proof

Proof by contradiction; Let $s^+ \in \mathbb{P}, \hat{D}; s^+ \in \mathbb{P} \cap \hat{D}$

$$X_i = \{x_1, ..., x_n\}$$
$$D := f[X_i] \to a_o \in \{\mathbb{T}, \mathbb{F}\} \;\;\; \forall X_i$$

$$Let \;\; D \in \mathbb{P}$$
$$s^+ := P[X_i] \to y_o : y_o = a_o \;\;\; \forall X_i$$
$$O[n] = O_T[n] + O_S[n] < (\lambda_K n)^K + (\lambda_{K-1} n)^{K-1} ... + \lambda_1 n + C \;\;\; \forall n$$

Traveling Salesman Problem of Dimension 2

22 Proof of the existence of $\hat{\mathcal{D}}$

22.1 The Traveling Salesman Problem of Dimension 2

English description

22.2 Formal Definition

$$X_i = \{l_1, l_2, ..., l_n, C\}$$

$$l_i = \{x_i, y_i\} \ \ \forall i$$

l_i denotes the 2D coordinates of location i

$$C = \{d_{proposed}, p_{decimal}\}$$

$d_{proposed}$ denotes the suggested shortest distance

$p_{decimal}$ is the decimal precision

$$L[l_i, l_j] := \sqrt{(y_j - y_i)^2 + (x_j - x_i)^2}$$

Let $L[l_i, l_j]$ denote the distance between location l_i and l_j

$$\tilde{L}[l_i, l_j] := d_{trunc} : -p_{decimal} < d_{trunc} - L[l_i, l_j] < p_{decimal}$$

Let $\tilde{L}[l_i, l_j]$ denote a truncated decimal representation of $L[l_i, l_j]$

$$R_i := \{r_1, r_2, ..., r_n, r_1\} \ \ : \ \ r_i \in X_i \ \ \forall i; \ \ r_i \neq r_j$$

Let R_i denote route i

$$L_{Total}[R_i] := (\textstyle\sum_{i=1}^{n-1} \tilde{L}[r_i, r_{i+1}]) + \tilde{L}[r_n, r_1]$$

Let $L_{Total}[R_i]$ denote the sum of truncated lengths of route R_i

$$D := f[X_i] \rightarrow a_o \in \{\mathbb{T}, \mathbb{F}\} \ \ \forall X_i$$

$$a_o =$$

$$(\exists R_k : L_{total}[R_k] == d_{proposed}) \cap (\nexists R_j : L_{total}[R_j] < d_{proposed})$$

Traveling Salesman Problem of Dimension 2

22.3 Define subpath, subpath distance, subpath storage

$\tilde{L}[l_i, l_j]$ denotes "the distance of a subpath of length 1"

$$\tilde{L}[l_i, l_j] := d_{trunc} : -p_{decimal} < d_{trunc} - L[l_i, l_j] < p_{decimal}$$
$$= abs(d_{trunc} - L[l_i, l_j]) < p_{decimal}$$

$\tilde{R}$ denotes a subpath of length k

$$\tilde{R} = \{\tilde{r}_1, \tilde{r}_2, ..., \tilde{r}_k\} \ : \ \tilde{r}_i \in X_i \ \ \forall i, r_i \neq r_j$$

$\tilde{L}_k[\tilde{R}]$ denotes "the distance of a subpath of length k"

$$\tilde{L}_k[\tilde{R}] := \sum_{i=1}^{k} \tilde{L}[\tilde{r}, \tilde{r}_{i+1}]$$

Let $\mathcal{M}_1$ denote the memory reserved for subpaths distances of length 1

$$\mathcal{M}_1 = \{\hat{b}_{1;1}, \hat{b}_{1;2}, \hat{b}_{1;3}, ..., \hat{b}_{startindex;finishindex}, ..., \hat{b}_{n-1;n}\}^*$$
$$\mathcal{M} \supseteq \mathcal{M}_1$$

* Note $\hat{b}_{i;j} = \hat{b}_{j;i}$
$\sqrt{(y_j - y_i)^2 + (x_j - x_i)^2} = \sqrt{(y_i - y_j)^2 + (x_i - x_j)^2}$

22.4 Define the following functions

22.4.1 $sqrt[x, p_{decimal}] = \sqrt{x}$ [1]

22.4.2 $pow[x, 2, p_{decimal}] = x^2$ [2]

22.5 Define the following subfunctions

22.5.1 loadM1Subpaths[X]

// Computes all subpaths of length 1 and stores in $\mathcal{M}_1 = \{\hat{b}_{1;1}, \hat{b}_{1,;2}, ..., \hat{b}_{n-1;n}\}$

$$//X_i = \{l_1, l_2, ..., l_n, C\}$$
$$//l_i = \{x_i, y_i\} \ \ \forall i$$

$$//\mathcal{M} = \{b_1, b_2, ..., b_M, \hat{b}_{1;1}, \hat{b}_{1,;2}, ..., \hat{b}_{n-1;n}, y_o\} = \{b_1, b_2, ..., b_M, \mathcal{M}_1, y_o\} = \{\mathcal{M}, \mathcal{M}_1, y_o\}$$

$$\forall i, j > i$$

$$b_3 \leftarrow y_i - y_j$$

$$b_4 \leftarrow x_i - x_j$$

$$b_3 \leftarrow b_3^2$$

$$b_4 \leftarrow b_4^2$$

$$b_3 \leftarrow b_3 + b_4$$

$$\hat{b}_{i;j} \leftarrow \sqrt{b_3}^*$$

$^*\hat{b}_{i;j} = \tilde{L}[l_i, l_j]$

22.5.2 computeAllRoutes$[X]$

// Computes all complete routes, checks for a route == $d_{proposed}$, sets y_o to false if the current route is shorter than $d_{proposed}$

$$\forall i, j \neq i, k \neq i, j..., q \neq i, j, ..., m$$

$$b_3 \leftarrow \hat{b}_{1;j} + \hat{b}_{j;k}$$

$$b_3 \leftarrow b_3 + \hat{b}_{k;l}$$

$$...$$

$$b_3 \leftarrow b_3 + \hat{b}_{m;q}$$

$$b_3 \leftarrow b_3 + \hat{b}_{q;1}$$

$$b_4 \leftarrow b_3 == b_2$$

$$b_1 \leftarrow b_1 \vee b_4$$

$$b_4 \leftarrow b_2 \leqslant b_3$$

$$y_o \leftarrow y_o \wedge b_4$$

22.6 Express a solution using subfunctions, storing subpaths of length 1 in memory

// $d_{proposed}$ is the shortest path

$$y_o \leftarrow \mathbb{T}$$

$$//d_{proposed} \text{ exists as a total path length}$$
$$b_1 \leftarrow \mathbb{F}$$

$$// \text{ shortest path register}$$
$$b_2 \leftarrow d_{proposed}$$

$$loadM1Subpaths[X]$$
$$computeAllRoutes[X]$$

22.7 Show each subfunction satisfies the subfunction condition of solutions : $P[\hat{X}_i] \supseteq P[X_i] \;\; \forall \hat{X}_i, X_i, \;\; \hat{X}_i \supseteq X_i$

Let

$$\mathcal{M}_0 = \{b_1, b_2, b_3, b_4, y_o\}$$
$$\mathcal{M}_1 = \{\hat{b}_{1;1}, \hat{b}_{1,;2}, ..., \hat{b}_{n-1;n}\}$$

22.7.1 $loadM1Subpaths[X] \rightarrow \mathcal{M}_1$

Let

$$//X = \{l_1, l_2, ..., l_n, C\}; \;\; \hat{X} = \{l_1, l_2, ..., l_n, l_{n+1}, C\}$$

$$loadM1Subpaths[X, \mathcal{M}] \rightarrow \mathcal{M}_1 = Sub_1[X, \mathcal{M}] \rightarrow \mathcal{M}_1$$

$$Sub_1[X, \mathcal{M}] = \{\mathcal{L}, \mathcal{M}\}$$
$$= \{\hat{b}_{i;j} \leftarrow \tilde{L}[l_i, l_j] \;\; \forall i, j > i | b_3, b_4, \hat{b}_{1;1}, \hat{b}_{1;2}, ..., \hat{b}_{n-1;n}\}$$
$$Sub_1[X_i, \mathcal{M}] = \{\hat{b}_{i;j} \leftarrow \tilde{L}[l_i, l_j] \;\; \forall i, j > i | b_3, b_4, \mathcal{M}_1\}$$

$$Sub_1[\hat{X}, \mathcal{M}] = \{\hat{\mathcal{L}}, \hat{\mathcal{M}}\}$$
$$= \{\hat{b}_{i;j} \leftarrow \tilde{L}[l_i, l_j] \;\; \forall i, j > i | b_3, b_4, \hat{b}_{1;1}, \hat{b}_{1;2}, ..., \hat{b}_{n;n+1}\}$$
$$= \{\mathcal{L}, \hat{b}_{i;j} \leftarrow \tilde{L}[l_i, l_j] \;\; \forall i, j = n+1 | \mathcal{M}, \hat{b}_{1;n+1}, \hat{b}_{2;n+1}, ..., \hat{b}_{n;n+1}\}$$
$$Sub_1[\hat{X}, \mathcal{M}] = \{\mathcal{L}, \mathcal{L}_{n+1} | \mathcal{M}, \mathcal{M}_{n+1}\}$$
$$Sub_1[\hat{X}, \mathcal{M}] = \{\mathcal{L}, \mathcal{L}_{n+1} | \mathcal{M}, \mathcal{M}_{n+1}\} \supseteq \{\mathcal{L} | \mathcal{M}\} = Sub_1[X, \mathcal{M}]$$

22.7.2 $computeAllRoutes[X]$

Let

$$computeAllRoutes[X] = Sub_2[X]$$
$$Sub_2[X] = \{\mathcal{L}, \mathcal{M}, y_o\}$$
$$= \{\hat{b}_{1;i_2} + \hat{b}_{i_2;i_3} + \hat{b}_{i_3;i_4} + ... + \hat{b}_{i_n;1} \;\; \forall i_2, i_3 \neq i_2, i_4 \neq i_2, i_3 ... i_n \neq i_2, i_3 ..., i_{n-1}$$
$$|b_1, b_2, b_3, b_4, \mathcal{M}_1, y_o\}$$

$$Sub_2[\hat{X}] = \{\hat{\mathcal{L}}, \hat{\mathcal{M}}, y_o\}$$
$$= \{\hat{b}_{1;i_2} + \hat{b}_{i_2;i_3} + \hat{b}_{i_3;i_4} + ... + \hat{b}_{i_{n+1};1} \;\; \forall i_2, i_3 \neq i_2, i_4 \neq i_2, i_3 ... i_{n+1} \neq i_2, i_3 ..., i_n$$
$$|\mathcal{M}, \mathcal{M}_{n+1}, y_o\}$$

Let

$$insert_subpath[\mathcal{L}] =$$

$$Sub_2[\hat{X}] = \{insert_subpath[\mathcal{L}, \hat{b}_{i_{n+1};j}, j] \;\; \forall j \neq n+1 | \mathcal{M}, \mathcal{M}_{n+1}, y_o\}$$

22.7.3 Show the overall solution storing subpaths of length 1 satisfies the subfunction condition of solutions : $P[\hat{X}_i] \supseteq P[X_i] \;\; \forall \hat{X}_i, X_i$

22.8 Express $O[n]$ in terms of subfunction complexities

$$O_{sub1}[n] = O_{T_{sub1}}[n] + O_{S_{sub1}}[n]$$
$$O_{sub2}[n] = O_{T_{sub2}}[n] + O_{S_{sub2}}[n]$$
$$O[n] = O_{sub1}[n] + O_{sub2}[n] + 3$$

22.9 $\mathbf{Sub_+[X]}$

Let $Sub_+[X]$ denote a subfuction that adds all subpaths of length 1
Let $O_+[n]$ denote the total complexity of subfunction $Sub_+[X]$

22.9.1 Find an expression for $O_+[n]$:= the number of $\tilde{L}[l_i, l_j] + \tilde{L}[l_j, l_k]$ length 1 subpath additions

$$O_+[n] = (\sum_{i=1}^{n} 1) \frac{(nP(n-1)}{2}$$

$$O_+[n] = \frac{n(n-1)!}{2}$$
$$O_+[n] = \frac{n!}{2}$$

22.10 Prove $Sub_+[X]$ is a subfunction of all s^+ by contradiction

suppose not all subpaths are considered
there could exist subpath resulting in an incorrect solution
contradicts definition of solution

22.11 Show the solution storing subpaths of length 1 contains $\mathbf{Sub_+[X]}$

$$// \ d_{proposed} \text{ is the shortest path}$$
$$y_o \leftarrow \mathbb{T}$$

$$//d_{proposed} \text{ exists as a total path length}$$
$$b_1 \leftarrow \mathbb{F}$$

$$// \text{ shortest path register}$$
$$b_2 \leftarrow d_{proposed}$$

$$loadM1Subpaths[X]$$
$$computeAllRoutes[X]$$

22.12 Express $O[n]$ in terms of subfunction complexities including $O_+[n]$ as a subfunction complexity

$$O[n] = O_{sub1}[n] + O_{sub2}[n] + 3$$
$$O_{sub2}[n] = O_+[n] + 8\frac{nP(n-1)}{2}$$
$$O_{sub2}[n] = O_+[n] + 8\frac{(n-1)!}{2}$$
$$O[n] = O_{sub1}[n] + O_+[n] + 8\frac{(n-1)!}{2} + 3$$

22.13 Show $limit_{n\to\infty}\frac{O_+[n+1]}{O_+[n]}$ diverges

$$limit_{n\to\infty}\frac{O_+[n+1]}{O_+[n]}$$

$$= limit_{n\to\infty} \frac{(n+1)!}{2} \frac{2}{n!}$$

$$= limit_{n\to\infty}\ n$$

There does not exist ...
therefore $limit_{n\to\infty} \frac{O_+[n+1]}{O_+[n]}$ diverges

22.14 Prove D diverges by the theorem of divergent subfunctions

22.15 Connection to "P $\neq$ NP"

Citations

[1] *chatgpt*
[2] $https://stackoverflow.com/questions/3518973/floating-point-exponentiation-without-power-function$
[3] $https://stackoverflow.com/questions/27086195/linear-index-upper-triangular-matrix$
[4] $http://www.math.uchicago.edu/\ may/VIGRE/VIGRE2011/REUPapers/Riffer-Reinert.pdf$